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Mannino et al.

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(54) **COCHLEATES MADE WITH SOY
PHOSPHATIDYLSERINE**

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A61K 31/7036 (2013.01); *A61K 31/7048*
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(57) **ABSTRACT**

Unpurified or low pure soy phosphatidylserine is used to
make cochleates. The cochleates contain about 40-74% soy
phosphatidylserine, a multivalent cation and a biological
active. A preferred cochleate contains the antifungal agent
amphotericin B.

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AMIKACIN COCHLEATE IN VITRO ACTIVITY

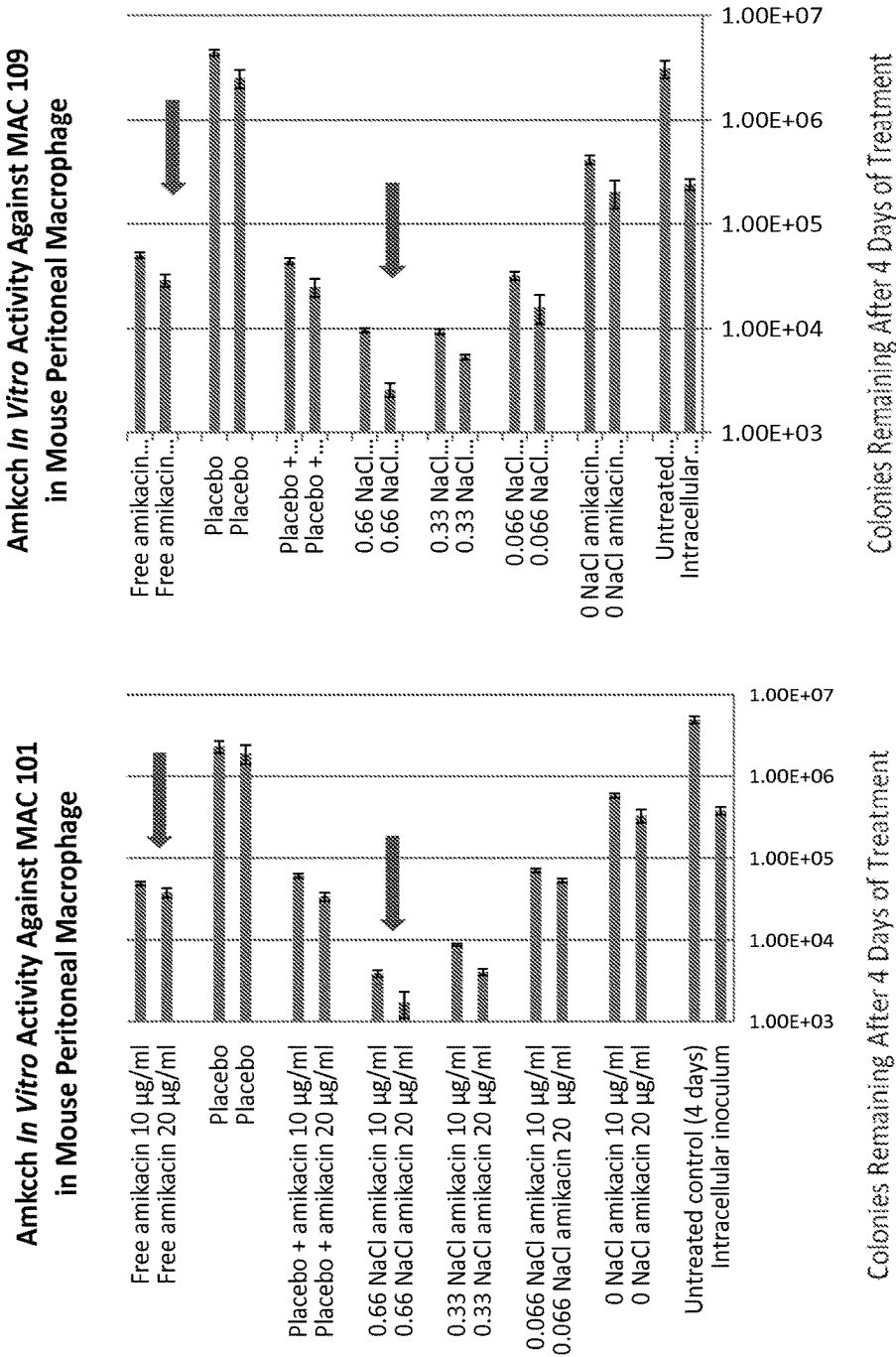


Figure 1

**AMKCCH /N VIVO EFFICACY IN A MOUSE MODEL OF
MAC 101 INFECTION – IP DELIVERY**

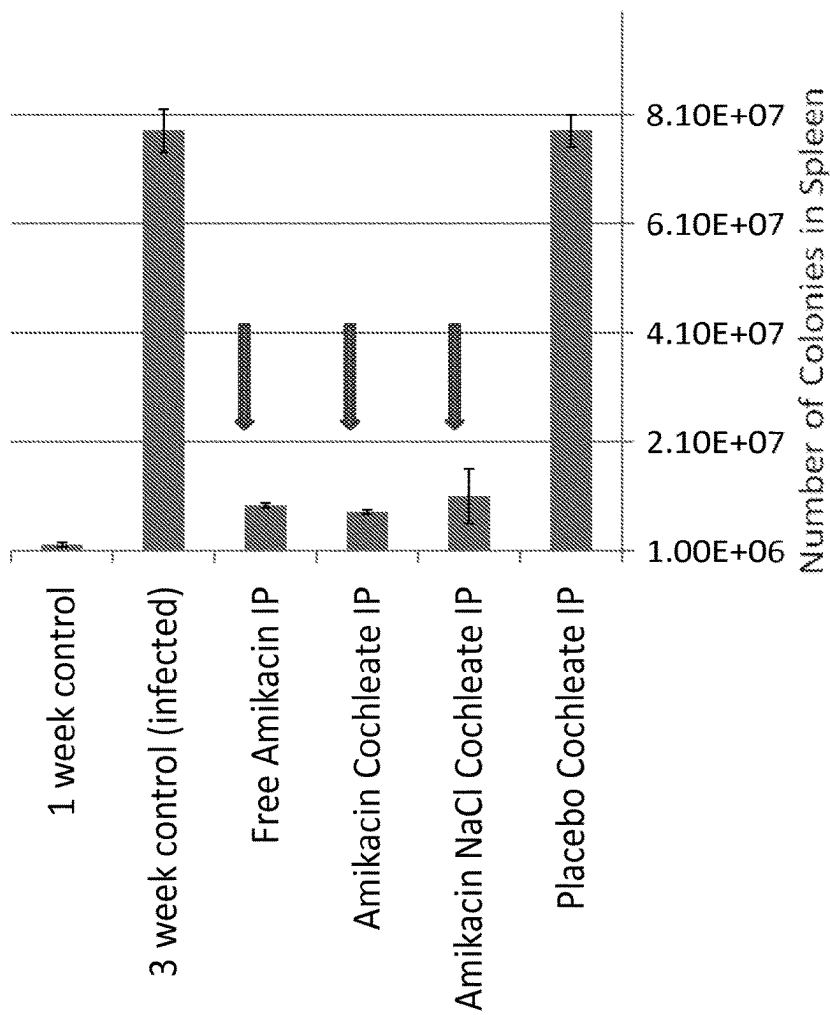


Figure 2

**AMKCCH *IN VIVO* EFFICACY IN A MOUSE MODEL OF
MAC 101 INFECTION – ORAL DELIVERY**

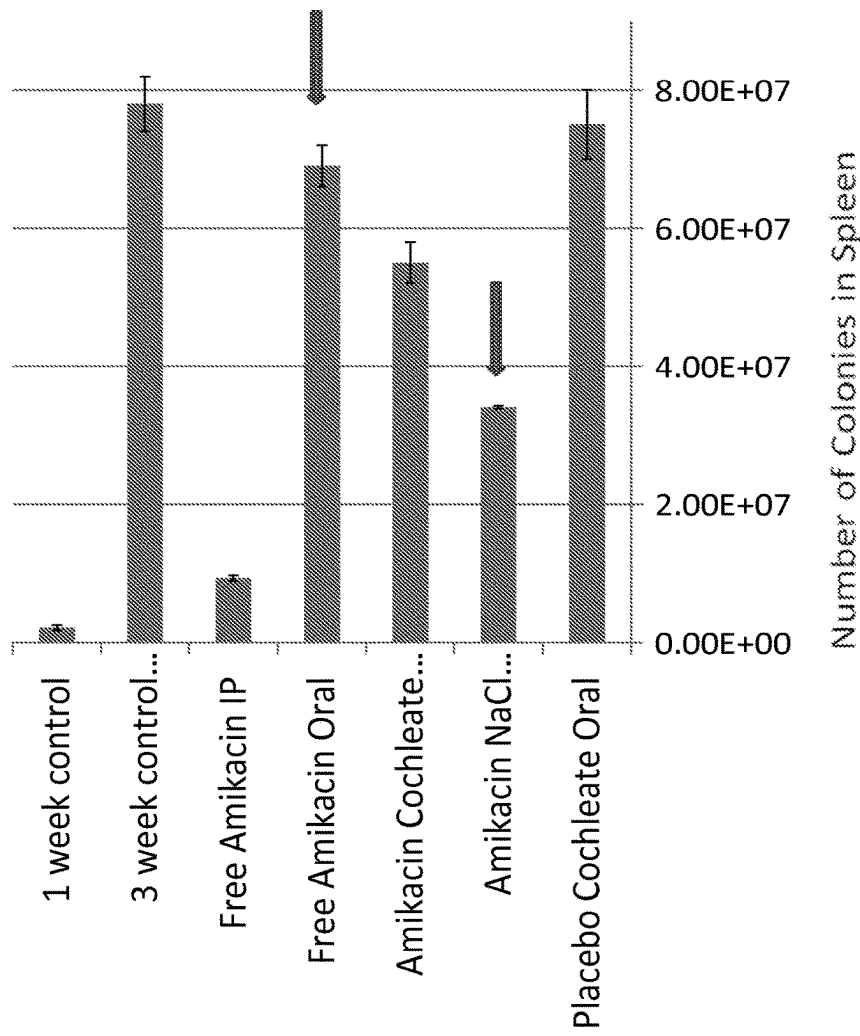
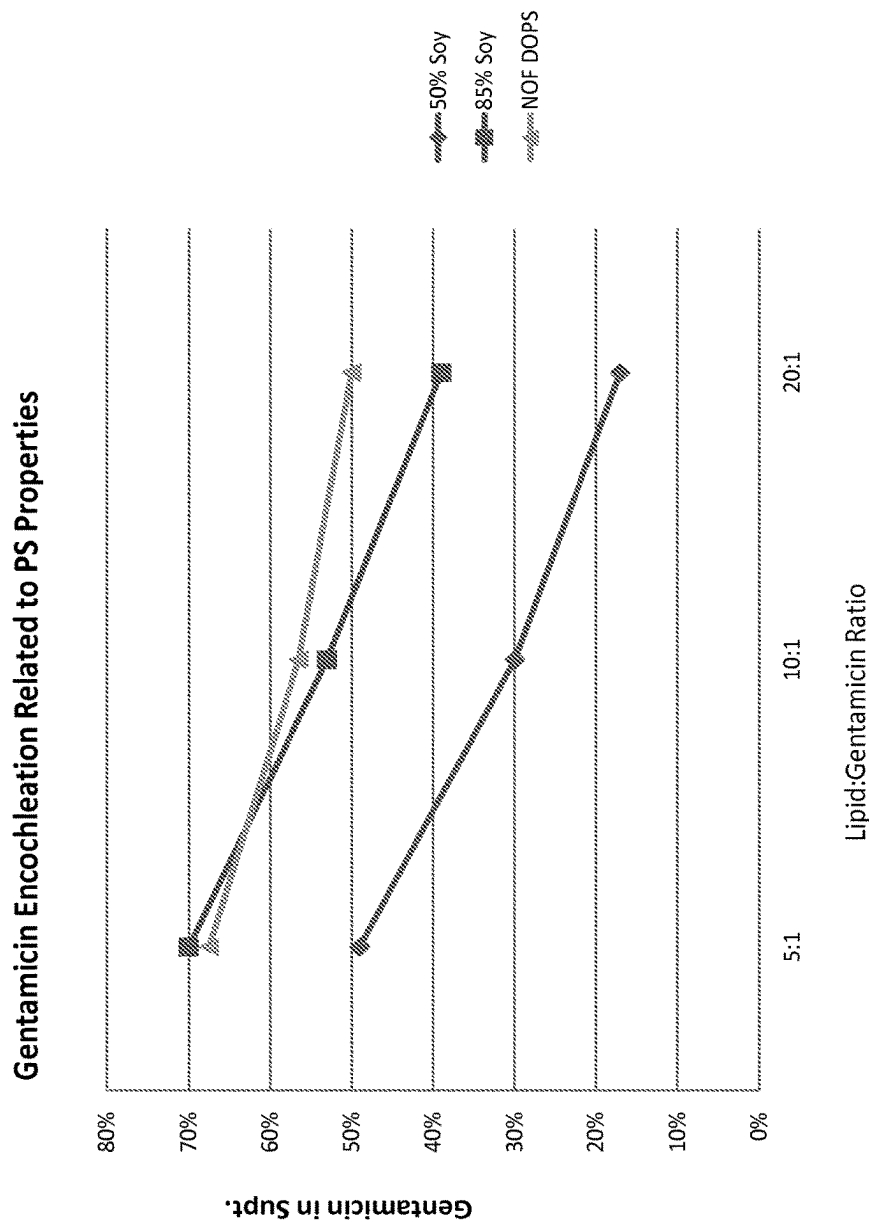
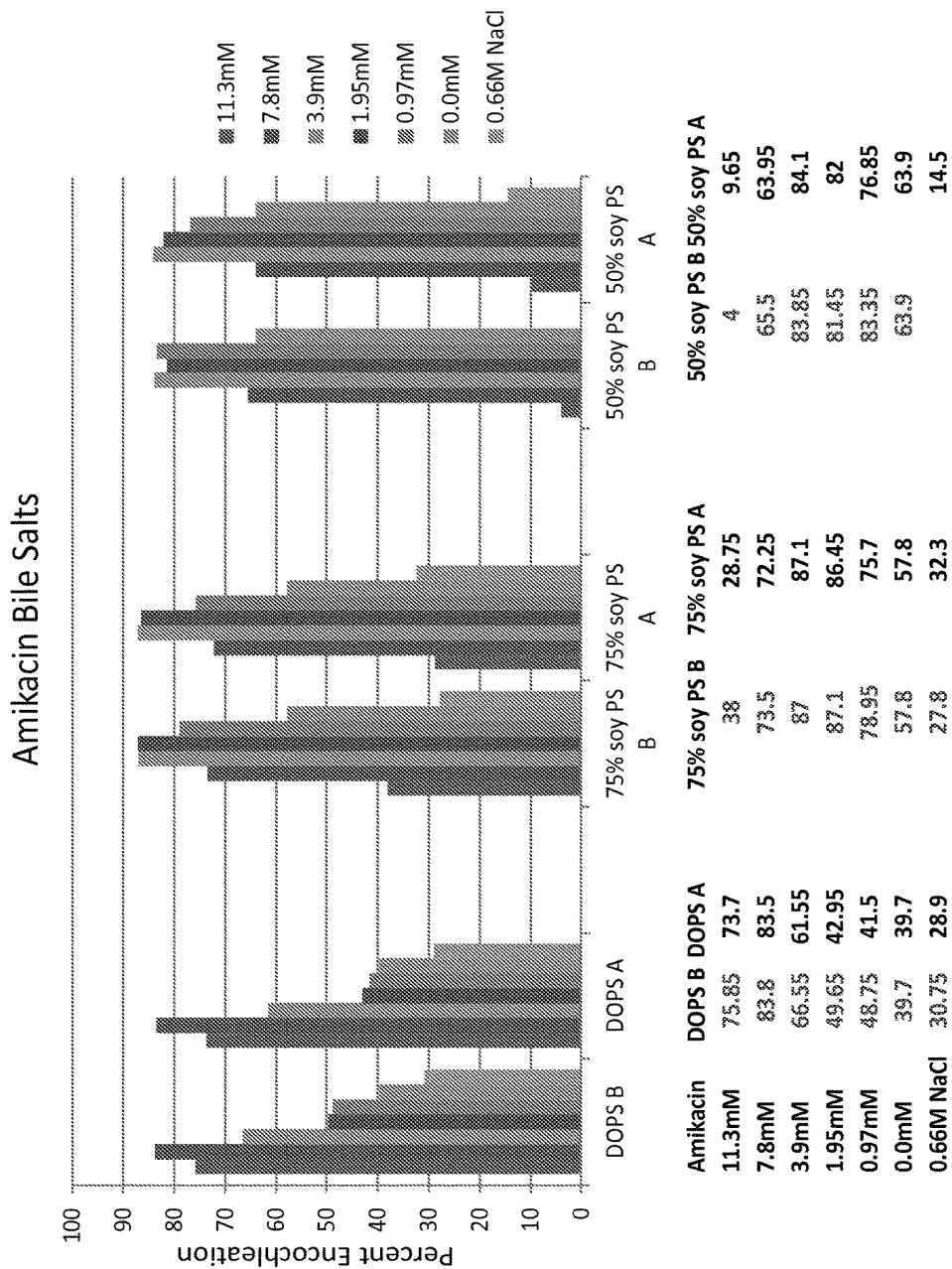


Figure 3



The efficiency of encochleation is the reciprocal of the percentage in the supernatant

Figure 4



A is bile salts added after the encocapsulation. B is bile salts added before the encocapsulation

Figure 5

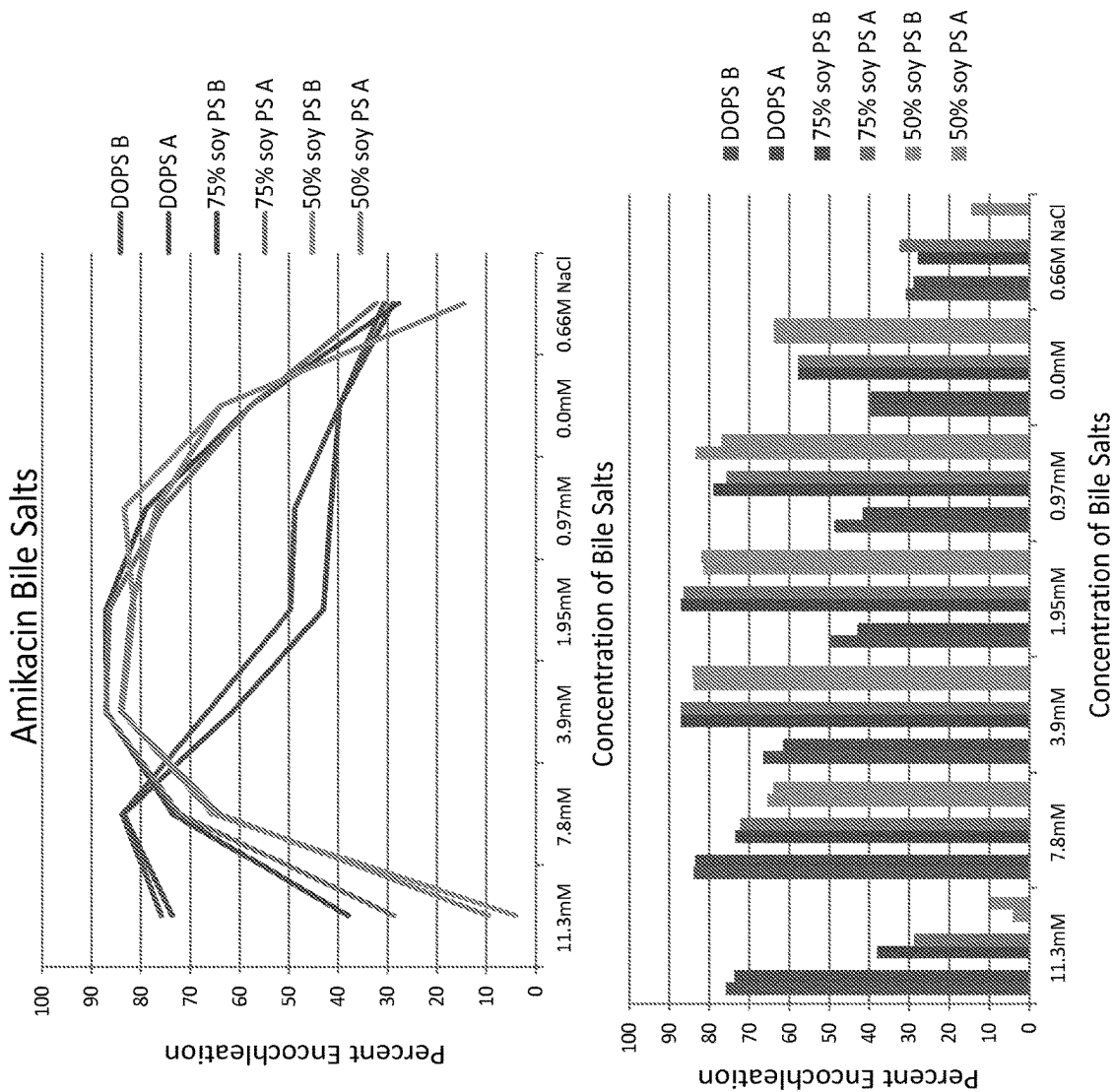


Figure 6

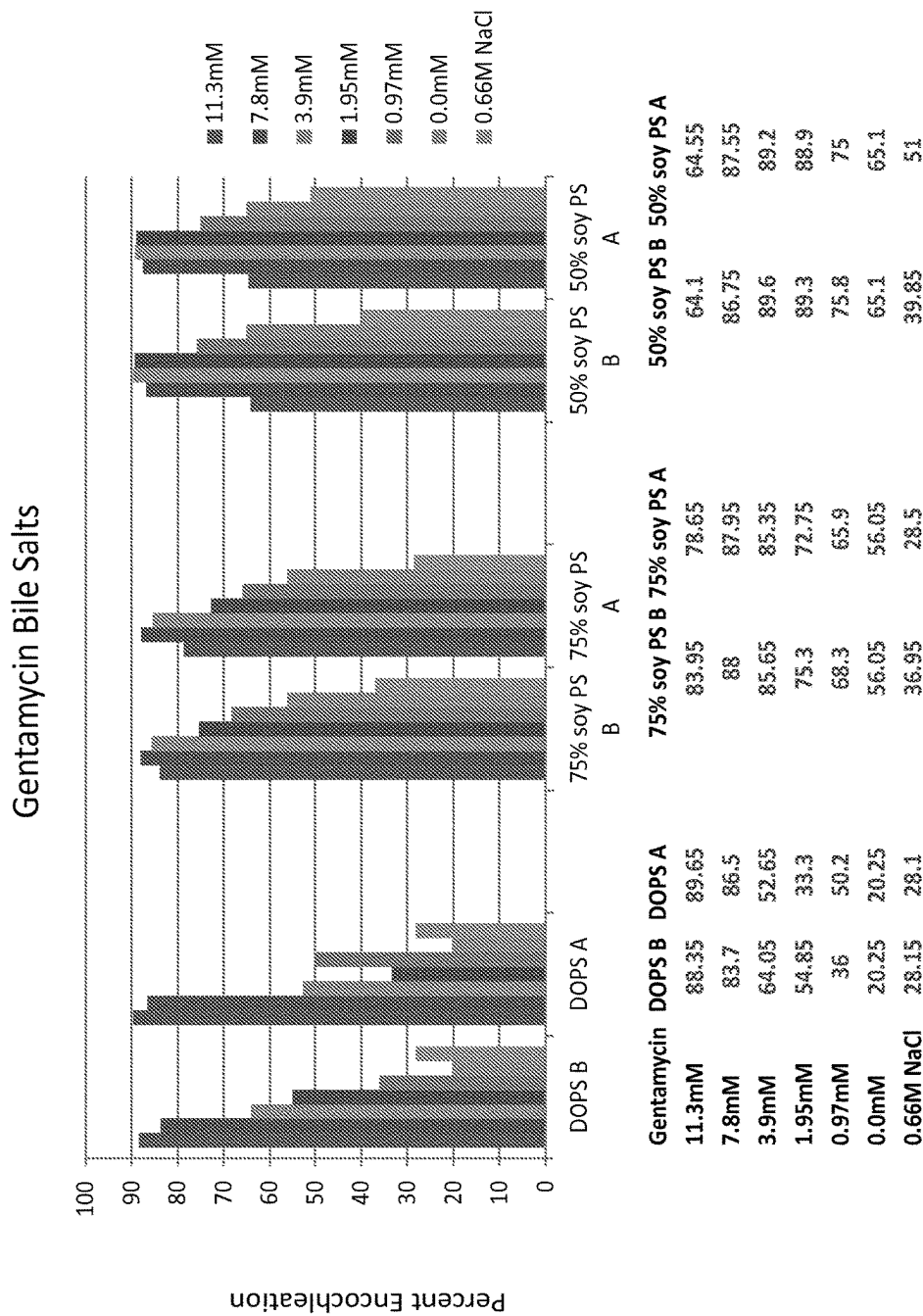


Figure 7

Gentamycin Bile Salts

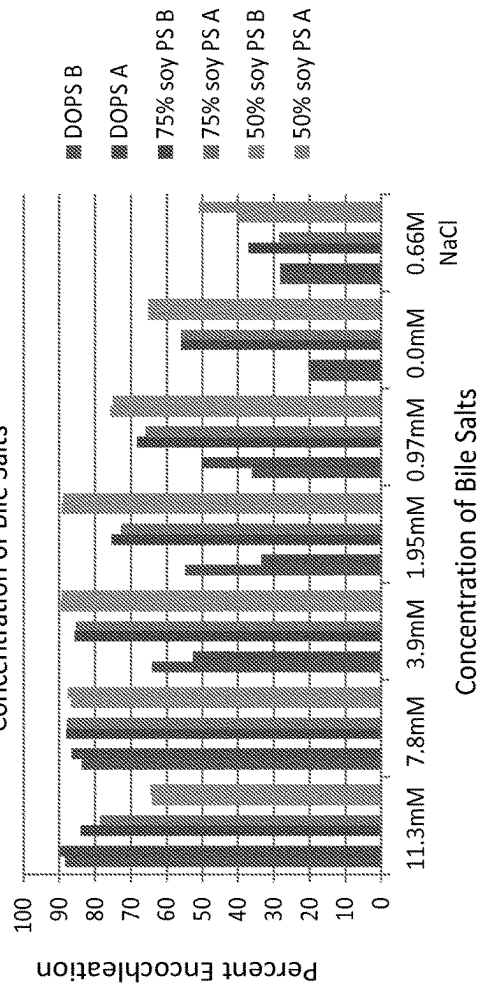
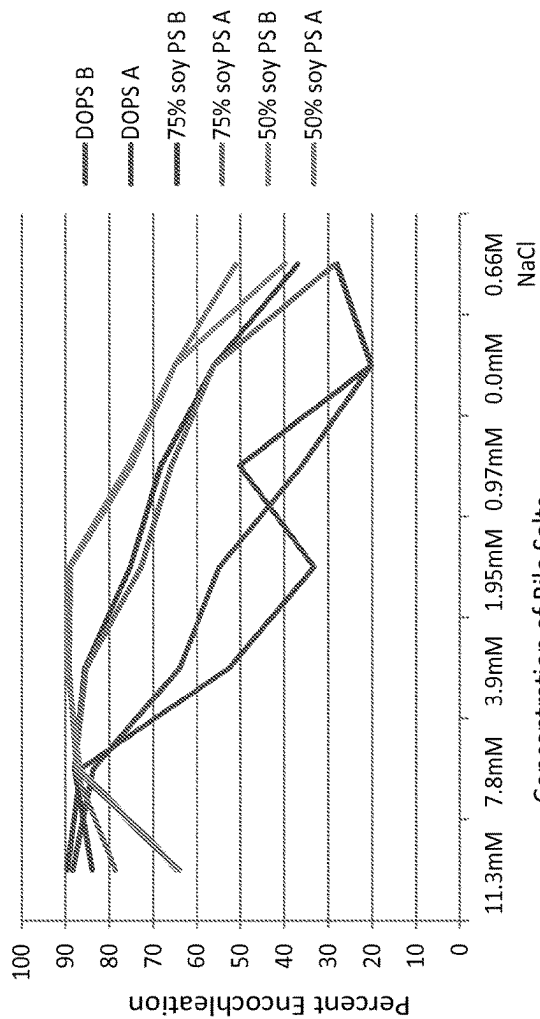
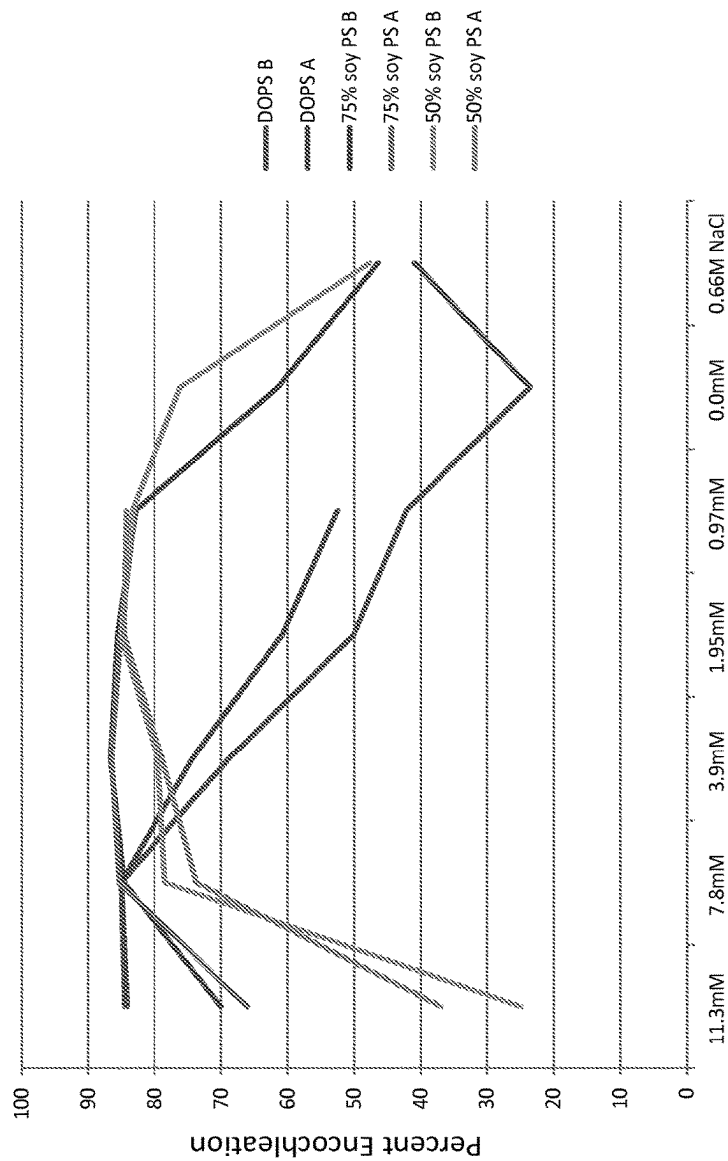


Figure 8

Paromomycin Bile Salts



Concentration of Bile Salts

Figure 9

Paromomycin Bile Salts

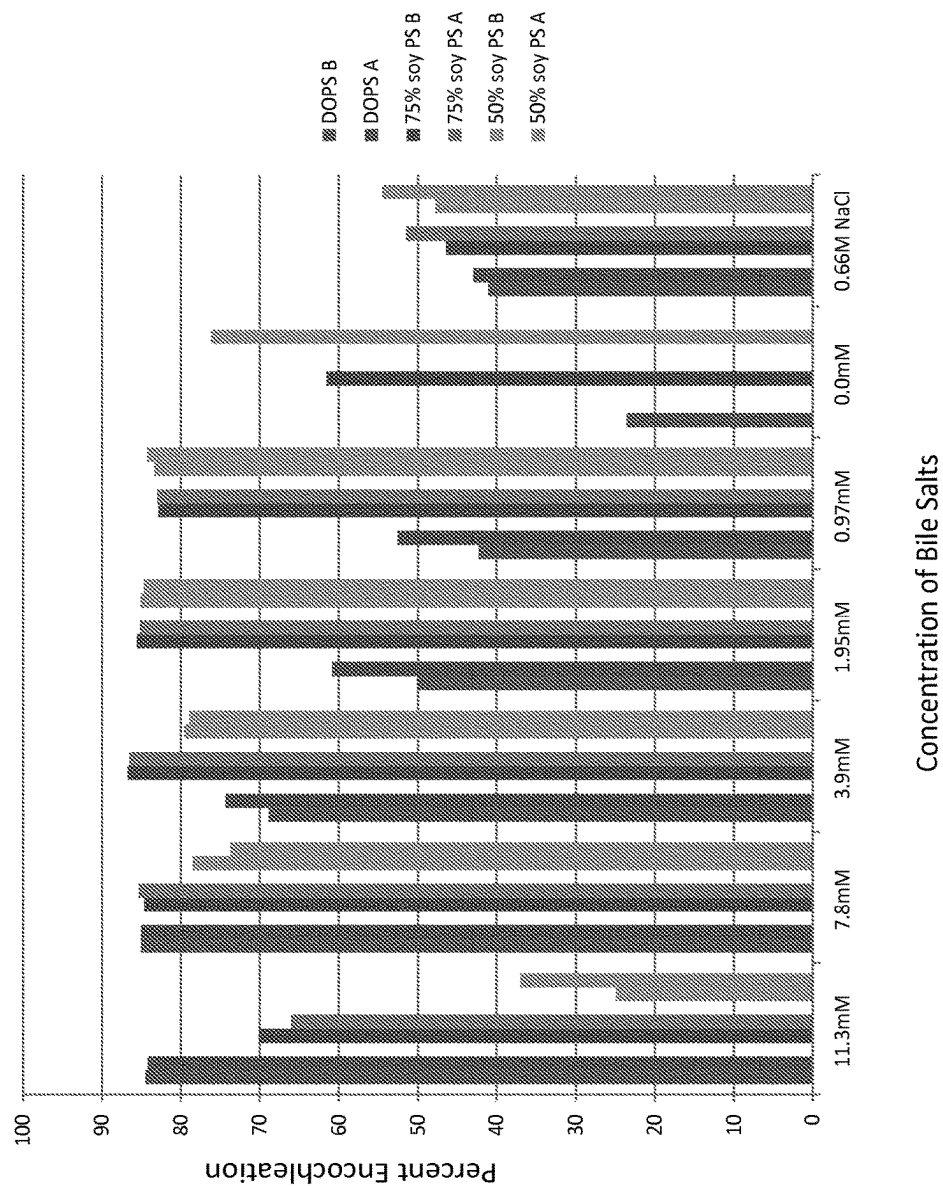


Figure 10

Paromomycin Bile Salts

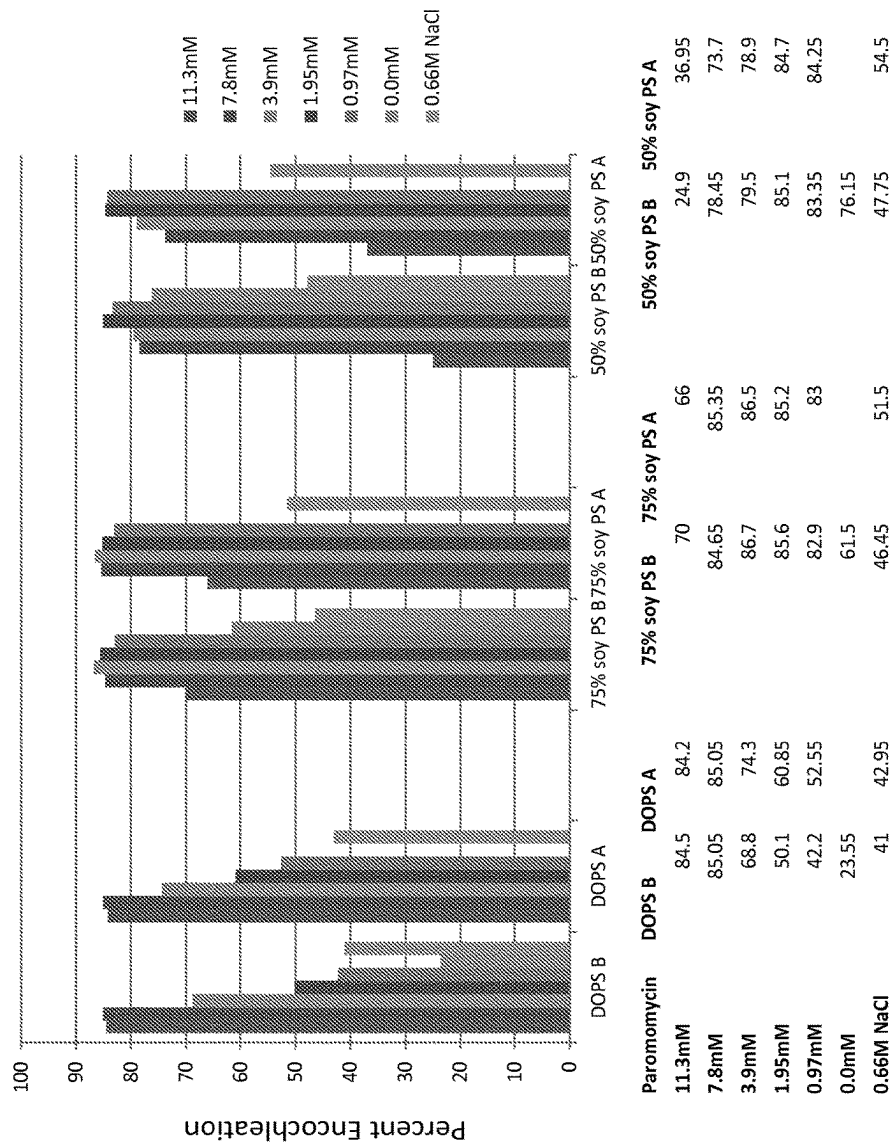


Figure 11

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COCHLEATES MADE WITH SOY PHOSPHATIDYLSERINE

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/904,824, filed on Jun. 18, 2020, which is a continuation of U.S. application Ser. No. 15/811,040, filed on Nov. 13, 2017, now U.S. Pat. No. 10,716,860, issued Jul. 21, 2020, which is a continuation of U.S. application Ser. No. 14/418,392, filed on Jan. 29, 2015, now abandoned, which is a 371 national stage application of International Application No. PCT/US13/52756, filed Jul. 30, 2013, which claims the benefit of U.S. Provisional Application No. 61/835,825, filed on Jun. 17, 2013, and U.S. Provisional Application No. 61/677,414, filed on Jul. 30, 2012. The entire contents of these applications are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

The present invention relates to the ability of unpurified or low purity (40-74% by weight) soy-based phosphatidylserine (PS) to prepare cochleates, methods of preparing drug-cochleates from soy-based PS and the use of this drug-loaded cochleate as a pharmaceutical treatment.

BACKGROUND OF THE INVENTION

Cochleate delivery vehicles are a broad-based technology for the delivery of a wide range of bioactive therapeutic products. Cochleate delivery vehicles are stable phospholipid-cation precipitates composed of simple, naturally occurring materials, for example, phosphatidylserine and calcium.

The bilayer structure of cochleates provides protection from degradation for associated, or "encochleated," molecules. Since the entire cochleate structure is a series of solid layers, components within the interior of the cochleate structure remain substantially intact, even though the outer layers of the cochleate may be exposed to harsh environmental conditions or enzymes. This includes protection from digestion in the stomach.

Taking advantage of these unique properties, cochleates have been used to mediate and enhance the oral bioavailability of a broad spectrum of important but difficult to formulate biopharmaceuticals, including compounds with poor water solubility, protein and peptide drugs, and large hydrophilic molecules. For example cochleate-mediated oral delivery of amphotericin B, large DNA constructs/plasmids for DNA vaccines and gene therapy, peptide formulations, and antibiotics such as clofazimine has been achieved.

Cochleates can be stored in cation-containing buffer, or lyophilized to a powder, stored at room temperature, and reconstituted with liquid prior to administration. Lyophilization has no adverse effects on cochleate morphology or functions. Cochleate preparations have been shown to be stable for more than two years at 4° C. in a cation-containing buffer, and at least one year as a lyophilized powder at room temperature.

Cochleates can be prepared by several methods, such as trapping or hydrogel methods (International Application Publication No. WO 03/082209, the entire content of which is incorporated herein by reference).

Soy PS is sold in health food stores as a nutritional supplement. Non-purified (40%) PS has been used and

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studied as a nutritional supplement and as a component that has a beneficial effect on enhancing the brain functions in elderly people (Villardita C et al., *Clin. Trials J.* 24, 1987, 84-93).

Although non-purified soy PS (NSPS) has been sold and studied on patients, NSPS (or low purity PS) has never been used to make cochleates and to deliver a drug using these cochleates. As previously disclosed in WO 03/082209, NSPS does not form cochleates and that a purification process is needed to enhance the NSPS in the content of PS, until at least about 75% by weight of PS is reached, such percentage allowing the formation of cochleates.

SUMMARY OF THE INVENTION

It has been unexpectedly found that NSPS or low purity soy-based PS (40-74% by weight) can still form cochleates.

Briefly, in accordance with the present invention, improved lipid based cochleates are made by using non-purified or low purity soy phosphatidylserine as the lipid source. The improved cochleates contain soy phosphatidylserine in an amount of about 40%-74% (preferably 45-70%, more preferably 45-55%) by weight of the lipid. The improved cochleates can be empty or loaded cochleates. Loaded cochleates can contain any biological actives or combination of biological actives such as, for example, a protein, a small peptide, a polynucleotide, an aminoglycoside, an antiviral agent, an anesthetic, an antibiotic, an antifungal, an anticancer, an immunosuppressant, a steroidal anti-inflammatory, a non-steroidal anti-inflammatory, a tranquilizer, a nutritional supplement, an herbal product, a vitamin or a vasodilatory agent. Of particular interest in practicing the present invention, antifungal agents or antibiotic agents are loaded into the present soy-based phosphatidylserine cochleates to provide a cost effective and improved antifungal drug/antibiotic drug with reduced toxicity. Preferred antifungal agents include amphotericin-B and nystatin. Preferred antibiotic agents include an aminoglycoside and amikacin.

The improved lipid based cochleates of the present invention can be made by a method comprising the steps of: (a) preparing liposomes in an aqueous medium wherein the liposomes have (i) a lipid bilayer comprising soy-based phosphatidylserine in an amount of about 40%-74% (preferably 45-70%, more preferably 45-55%) by weight of the lipid bilayer and (ii) a load of a biological active; (b) adding a multivalent cation to the suspension of liposomes of (a) to form the soy phosphatidylserine/biological active cochleates; and (c) collecting the soy-based phosphatidylserine/biological active cochleates.

The present invention also teaches that the soy phosphatidylserine/biological active cochleates can be administered to patients with fungal infections or with bacterial infections. The present soy phosphatidylserine/biological active cochleates are conveniently administered orally even in the treatment of systemic fungal infections of immune compromised patients. The present phosphatidylserine/biological active cochleates are also administered parenterally, or by other means of administration. The preferred biological active is amphotericin-B, curcumin, and amikacin.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows amikacin cochleate in vitro study against MAC 101 and MAC 109 in mouse peritoneal macrophage.

FIG. 2 shows amikacin cochleate in vivo efficacy in a mouse model of MAC 101 infection IP-delivery.

FIG. 3 shows amikacin cochleate in vivo efficacy in a mouse model of MAC 101 infection oral-delivery.

FIG. 4 shows gentamicin encochleation related to PS properties.

FIG. 5 shows encochleation efficiency of amikacin bile salts related to PS properties.

FIG. 6 shows encochleation efficiency of amikacin bile salts related to PS properties.

FIG. 7 shows encochleation efficiency of gentamycin bile salts related to PS properties.

FIG. 8 shows encochleation efficiency of gentamycin bile salts related to PS properties.

FIG. 9 shows encochleation efficiency of paromomycin bile salts related to PS properties.

FIG. 10 shows encochleation efficiency of paromomycin bile salts related to PS properties.

FIG. 11 shows encochleation efficiency of paromomycin bile salts related to PS properties.

DETAILED DESCRIPTION OF THE INVENTION

The following terms when used herein will have the definitions given below.

A "cochleate" is a stable, phospholipid-cation precipitate that can be either empty or loaded.

An "empty cochleate" is a cochleate that is comprised only of phospholipid and cations.

A "loaded cochleate" is a cochleate that has one or more biological active compounds within the phospholipid-cation structure.

"Soy phosphatidylserine" or "soy-based phosphatidylserine" is phosphatidylserine that has been derived from a soy based composition.

In practicing the present invention improved phospholipid based cochleates are made by using soy phosphatidylserine in an amount of 40%-74% by weight of the lipid component of the cochleates. Alternatively, the soy phosphatidylserine can be about 40%, 45%, 50%, 55%, 60%, 65%, or 70%, or any incremental value thereof, by weight of the lipid component of the cochleates. It is to be understood that all values, and ranges between these values and ranges are meant to be encompassed by the present invention. In a preferred embodiment the phospholipid comprises 45-70% soy phosphatidylserine. In a more preferred embodiment, the phospholipid comprises 45-55% soy phosphatidylserine.

Phosphatidic acid is a preferred phospholipid when there is an additional phospholipid besides phosphatidylserine in the presently improved cochleates. Other phospholipids in addition to phosphatidic acid that can be used in the presently improved cochleates include phosphatidylcholine, phosphatidylinositol and phosphatidylglycerol. Mixtures of the additional phospholipids can also be used in combination with the soy phosphatidylserine.

The soy phosphatidylserine starting material is commercially available, or can be purified from soy phospholipid composition, which are mixtures of several soy phospholipids, according to well known and standard purification techniques.

Any multivalent compound can be used to precipitate the cochleates from the liposome starting materials. Preferably, the multivalent compounds are divalent cations such as Ca^{++} , Zn^{++} , Ba^{++} , and Mg^{++} . Preferred sources of these cations include the chloride salts of calcium, zinc, barium, and magnesium. CaCl_2 is a particularly preferred source of divalent cations.

In one embodiment, the present soy phosphatidylserine cochleates may further comprise bile salts. The weight ratio of soy-based phospholipid to the bile salts is between 20:1 and 0.5:1, preferably, between 10:1 and 3:1.

Bile salts are bile acids compounded with a cation, usually sodium. Bile acids are steroid acids found predominantly in the bile of mammals. Bile salts are commercially available (for example, Sigma—Aldrich catalog #48305 Fluka cholic acid sodium salt, ~50% deoxycholic acid sodium salt, ~50%).

It has been unexpectedly found that the addition of bile salts enhances the encochleation efficiency of the soy phosphatidylserine cochleates. For example, with inclusion of bile salts, the soy phosphatidylserine cochleates of the present invention (e.g., with the use of 50% soy PS) are more efficient at encochleating than cochleates containing at least about 75% soy phosphatidylserine.

In one embodiment the present soy phosphatidylserine cochleates are made by a process which comprises the steps of:

- a. preparing liposomes in an aqueous medium wherein the liposomes have (i) a lipid bilayer comprising soy-based phosphatidylserine in an amount of about 40%-74% (preferably 45-70%, more preferably 45-55%) by weight of the lipid bilayer and (ii) a load of a biological active;
- b. adding a multivalent cation to the suspension of liposomes of (a) to form the soy phosphatidylserine/biological active cochleates; and
- c. collecting the soy-based phosphatidylserine/biological active cochleates.

In one embodiment, the aqueous medium containing the suspension of liposome is a buffered environment having a pH of 6.5-7.5, and the load of the biological active is at pH 10 or higher prior to addition to the liposomes. In a preferred embodiment, the suspension of liposomes is buffered with phosphate.

In another embodiment, the method further comprises a step of adding bile salts to the suspension of liposomes of (a) before step (b) or adding bile salts to the soy-based phosphatidylserine/biological active cochleates after step (b), wherein the weight ratio of the lipid bilayer to the bile salts is between 20:1 and 0.5:1, preferably, between 10:1 and 3:1.

The present invention provides a geodate composition which contains (1) a lipid monolayer including a soy-based phospholipid that comprises about 40%-74% (preferably 45-70%, more preferably 45-55%) by weight soy phosphatidylserine, disposed about a hydrophobic domain; and (2) a lipid strata disposed about the lipid monolayer, wherein the lipid strata comprises a structure of alternating cationic layers comprising a divalent cation and negatively charged lipid sheet-like layers; and a cargo moiety associated with the hydrophobic domain.

International Patent Application Publication No. WO 2004/041247 has disclosed a process to make a geodate composition, the entire content of which is incorporated herein by reference.

The bioactive active/drug (referred to as "load" or drug) can be hydrophobic in aqueous media, hydrophilic or amphiphilic. The drug can be, but is not limited to, a protein, a small peptide, a bioactive polynucleotide, an antifungal agent, an antiviral agent, an anesthetic, an anti-infectious agent, an antifungal agent, an anticancer agent, an immunosuppressant, a steroidal anti-inflammatory, a nutritional supplement, an herbal product, a vitamin, a non-steroidal anti-inflammatory, a tranquilizer or a vasodilatory agent. Examples include Amphotericin B, acyclovir, adriamycin,

vitamin A, cabamazepine, melphalan, nifedipine, indomethacin, naproxen, estrogens, testosterone, steroids, phenytoin, ergotamines, cannabinoids rapamycin, propanidid, propofol, alphadione, echinomycine, miconazole nitrate, teniposide, taxanes, paclitaxel, and taxotere.

The drug can be a polypeptide such as cyclosporin, angiotensin I, II and III, enkephalins and their analogs, ACTH, anti-inflammatory peptides I, II, III, bradykinin, calcitonin, b-endorphin, dinorphin, leucokinin, leutinizing hormone releasing hormone (LHRH), insulin, neurokinins, somatostatin, substance P, thyroid releasing hormone (TRH) and vasopressin.

The drug can be an antigen, but is not limited to a protein antigen. The antigen can also be a carbohydrate or DNA. Examples of antigenic proteins include envelope glycoproteins from influenza or Sendai viruses, animal cell membrane proteins, plant cell membrane proteins, bacterial membrane proteins and parasitic membrane proteins.

The antigen is extracted from the source particle, cell, tissue, or organism by known methods. Biological activity of the antigen need not be maintained. However, in some instances (e.g., where a protein has membrane fusion or ligand binding activity or a complex conformation which is recognized by the immune system), it is desirable to maintain the biological activity. In these instances, an extraction buffer containing a detergent which does not destroy the biological activity of the membrane protein is used. Suitable detergents include ionic detergents such as cholate salts, deoxycholate salts and the like or heterogeneous polyoxyethylene detergents such as Tween, BRIG or Triton.

Utilization of this method allows reconstitution of antigens, more specifically proteins, into the liposomes with retention of biological activities, and eventually efficient association with the cochleates. This avoids organic solvents, sonication, or extreme pH, temperature, or pressure all of which may have an adverse effect upon efficient reconstitution of the antigen in a biologically active form.

The presently improved cochleates can include loads with multiple antigenic molecules, biologically relevant molecules or drug formulations as appropriate.

To isolate the cochleate structures and to remove the polymer solution, cochleate precipitates are repeatedly washed with a buffer containing a positively charged molecule, and more preferably, a divalent cation. Addition of a positively charged molecule to the wash buffer ensures that the cochleate structures are maintained throughout the wash step, and that they remain as precipitates.

The medium in which the cochleates are suspended can contain salt such as sodium chloride, sodium sulfate, potassium sulfate, ammonium sulfate, magnesium sulfate, sodium carbonate. The medium can contain polymers such as Tween 80 or BRIG or Triton. The drug-cochleate is made by diluting into an appropriate pharmaceutically acceptable carrier (e.g., a divalent cation-containing buffer).

The cochleate particles can be enteric. The cochleate particles can be placed within gelatin capsules and the capsule can be enteric coated.

The skilled artisan can determine the most efficacious and therapeutic means for effecting treatment practicing the instant invention. Reference can also be made to any of numerous authorities and references including, for example, "Goodman & Gillman's, The Pharmaceutical Basis for Therapeutics", (6th Ed., Goodman et al., eds., MacMillan Publ. Co., New York, 1980).

The improved soy phosphatidylserine cochleates of the present invention containing a biological active are conveniently administered to patients orally whereby the cochle-

ates are absorbed into the bloodstream and the bioactive loads are delivered systemically. This is a particular advantage for water insoluble drugs such as amphotericin-B and paclitaxel. Additionally, the toxicity of many hydrophobic drugs is substantially reduced as seen with soy phosphatidylserine cochleates containing amphotericin-B as the load.

The following examples illustrate the practice of the present invention but should not be construed as limiting its scope.

Example 1 Amphotericin B Crystal Cochleate Formulation

Procedures of Amphotericin B Crystal Formulation in Bench Size:

#1.1 Experiment Formulation:

43 mg (actually 40 mg Amphotericin B based on the potency assay with the concentration of 0.932 mg/mg) of Amphotericin B in 1.33 mL 0.1N NaOH was mixed with 200 mg dioleoyl phosphatidylserine (DOPS, from Avanti or NOF Corporation) in 6.6 mL 50 mM phosphate buffer pH 7.4 (liposome was filtered through 5 μ m filter) to form liposomes containing the Amphotericin B. 29.3 mg calcium chloride was then added into the resultant mixture to form crystal cochleates. To make nice crystalline cochleates, the lipid:Amphotericin B weight ratio was set to about 5:1.

#1.2 Experiment Formulation:

18 mg Amphotericin B in 0.6 mL 0.1N NaOH was then mixed with 90 mg DOPS (from Avanti or NOF Corporation) in 3.0 mL 50 mM phosphate buffer pH 7.4 (liposome was filtered through 5 μ m, 8 μ m, and 4.5 μ m filter) to form liposomes containing the Amphotericin B. 0.33 mL 0.5M calcium chloride solution was then added into the resultant mixture to form crystals Amphotericin B cochleates. To make nice crystalline cochleates, the lipid:Amphotericin B weight ratio was about 5:1.

Procedures of Amphotericin B (Amphotericin B) Crystals Cochleates in Small Scale Size:

#1.3 Experiment Formulation:

399 mg (actually 372 mg based on the potency assay with the concentration of 0.932 mg/mg) Amphotericin B in 12.4 mL 0.1N NaOH solution was combined with 1.86 g of 50% soy PS (American Lecithin Company or Lipoid LLC) in 62 mL 50 mM phosphate buffer pH 7.4 (liposome was filtered through 5 μ m filter) to form liposomes containing the Amphotericin B. In order to add an antioxidant the suspension, 372 mg vitamin E in 3.72 mL ethyl alcohol was then added into the mixture of the liposomes. 520 mg powder of calcium chloride was then added into the resultant mixture to form Amphotericin B crystals cochleates. To make nice crystalline cochleates of Amphotericin B, the lipid:Amphotericin B weight ratio was about 5:1.

Procedures of Amphotericin B Crystal Formulation Scale Up to 4.5 L:

#1.4 Experiment Formulation:

22.75 g Amphotericin B (actually 21.2 g Amphotericin B based on the potency assay with the concentration of 0.932 mg/mg) in 707 mL 0.1N NaOH solution was then mixed with 106 g of 50% soy PS in 3.533 L 50 mM phosphate buffer pH 7.4 (liposome was filtered through 10 μ m filter) to form liposomes containing the Amphotericin B. To make a stable suspension of cochleates, 372 mg vitamin E in 3.72 mL ethyl alcohol was then added into the mixture. 194.8 mL 1M calcium chloride was then added into the resultant mixture to form crystal cochleates. The final product of cochleates was lyophilized with Freezer Dryer for a few

days. To make nice crystalline cochleates of Amphotericin B, the lipid:drug weight ratio was set to about 5:1.

Example 2 Curcumin (Granular) Geode Cochleates Formulation

Procedures of Curcumin (granular) Geode Cochleates in Small Scale Size:

#2.1 Experiment Formulation:

50 mg curcumin in 1 g castor oil and 4.0 mL ethyl alcohol was dissolved first then combined with 19 of 75% soy PS (American Lecithin Company or Lipoid LLC) in 100 mL sterile water (liposome was filtered through 5 μ m filter) to form liposomes containing the curcumin. In order to decrease the stickiness of suspension and batter diffusion, 500 mg bovine serum albumin ("BSA", or casein) was then added into the mixture of the liposome. 16.9 mL 0.1M calcium chloride was then added into the resultant mixture to form curcumin geode cochleates. The final product of geode cochleates was lyophilized with a freezer dryer for a few days. To make nice geode cochleates of curcumin, the lipid:drug weight ratio was set to about 20:1.

#2.2 Experiment Formulation:

100 mg 75% soy PS in 100 mg sterile water (liposome was filtered through 5 μ m filter) was mixed with 500 mg casein (or BSA) and then mixed with 50 mg curcumin in 2.0 g castor oil and 4.0 mL ethyl alcohol to form liposome containing the curcumin: 16.9 mL 0.1M calcium chloride was then added into the resultant mixture to form geode cochleates. The final product of geode cochleates was lyophilized with a freezer dryer for a few days. To make nice geode cochleates of curcumin, the lipid:drug weight ratio was set to about 20:1.

Procedures of Curcumin (Granular) Crystal Formulation in Small Scale Size:

#2.3 Experiment Formulation:

1000 mg 75% soy PS in 100 ml sterile water was (liposome was filtered through 5 μ m filter) combined with 20 mg curcumin in 4.0 mL ethyl alcohol to form liposome containing the curcumin. In order to help the drug stable in the lipid bilayer, 500 mg casein (or BSA) was then added into the suspension of the liposome. 16.9 mL 0.1M calcium chloride was then added into the resultant mixture to form crystal cochleates. The final product of crystal cochleates was lyophilized with Freezer Dryer for a few days. To make nice crystalline cochleates, the lipid:curcumin weight ratio was set to about 50:1.

#2.4 Experiment Formulation:

1000 mg 75% soy PS in 100 ml sterile water was (liposome was filtered through 5 μ m filter) combined with 20 mg curcumin in 4.0 mL ethyl alcohol to form liposome containing the curcumin. 16.9 mL 0.1M calcium chloride was then added into the resultant mixture to form crystal cochleates. In order to help the drug stable in the lipid bilayer, 500 mg casein (or BSA) was then added into the suspension of the cochleates. The final product of crystal cochleates was lyophilized with a Freezer Dryer for a few days. To make nice crystalline cochleates, the lipid:curcumin weight ratio was set to about 50:1.

Example 3 Amphotericin B (Amphotericin B) Geode Formulation

Procedures of Amphotericin B (Amphotericin B) Geode Formulation in Bench Size:

#3.1 Experiment Formulation:

7.5 mg Amphotericin B (actually 7 mg based on the potency assay with the concentration of 0.932 mg/mg) In 0.2 mL 0.1N NaOH solution was mixed with 50 mg of castor oil and then combined with 35 mg of 50% soy PS in 1.75 mL sterile water (liposome was filtered through 5 μ m filter) to form geode liposomes containing the Amphotericin B. To prevent the stickiness of Amphotericin B geode formulation, 21 mg BSA or casein was then added into the mixture of the Amphotericin B geode liposome. To increase the stability of the Amphotericin B geode formulation, 7 mg vitamin E in 70 μ L Ethyl alcohol was then added into the mixture of the geode liposome. To the resultant mixture, 127 μ L of 0.5M calcium chloride was then added to form Amphotericin B geode cochleates. To make a nice geode cochleates of Amphotericin B, the lipid:Amphotericin B weight ratio was about 5:1.

Procedures of Amphotericin B Geode Formulation Scale Up to 150 mL:

#3.2 Experiment Formulation:

536 mg Amphotericin B (actually 500 mg based on the potency assay with the concentration of 0.932 mg/mg) In 14.3 mL 0.1N NaOH solution was then mixed with 3.57 g castor oil and then combined with 2.5 g of 50% soy PS in 125 mL sterile water (liposome was filtered through 5 μ m filter) to form geode liposomes containing the Amphotericin B. To prevent the stickiness of the geode formulation, 1.5 g BSA or casein was then added into the mixture of geode liposome. To increase the stability of the Amphotericin B geode formulation, 250 mg of vitamin E in 2.5 mL ethyl alcohol was then added into the mixture of the geode liposome. To the resultant mixture was then added 8.88 mL of 0.5M calcium chloride to form geode cochleates. The pH of the final mixture was adjusted to neutral with 0.3 mL 1N HCl. To make nice geode cochleates of Amphotericin B, the lipid:Amphotericin B weight ratio was about 5:1.

#3.3 Experiment Formulation:

2.5 g of 50% soy PS in 125 mL sterile water (liposome was filtered through 5 μ m filter) was mixed with 1.5 g of BSA or casein and then combined with 536 mg of Amphotericin B (actually 500 mg based on the potency assay with the concentration of 0.932 mg/mg) in 14.3 mL 0.1N NaOH solution with 3.57 g of castor oil to form liposomes containing the Amphotericin B. To make a stable geode formulation, 250 mg vitamin E in 2.5 mL ethyl alcohol was then added into the mixture of geode liposome. The pH of the final mixture was adjusted to neutral with 0.3 mL 1N HCl. 8.88 mL of 0.5M calcium chloride was then added to the resultant mixture to form geode cochleates. Geode cochleates were then concentrated using lyophilization to provide geode cochleates with sterile water in any concentrations (based on the experiment requirements) with a lipid:Amphotericin B weight ratio of about 5:1.

Example 4 Amikacin Crystal Cochleate Formulation for In Vitro Study

Procedure of Amikacin Formulation for In Vitro Study:

#4.1 Experiment Formulation:

2 mg amikacin in 1.0 mL, distilled deionized (D.D) water was filtered through 0.22 μ m filter and combined with 20 mg of 50% soy PS in 2.0 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 0.206 mL of 0.1M calcium chloride to form cochleates. The mixture was then adjusting the drug concentration

of the amikacin at 0.5 mg/mL with sterile water. To make nice crystalline cochleates of amikacin with a lipid:drug ratio of about 10:1.

#4.2 Experiment Formulation:

2 mg amikacin in 1.0 mL D.D water was filtered through 0.22 μ m filter and combined with 20 mg of 50% soy PS in 2.0 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 0.206 mL of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 15 mg sodium chloride in 52 μ L sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the amikacin at 0.5 mg/mL with sterile water. Also, the final product containing the 0.066M sodium chloride. To make nice crystalline cochleates of amikacin with a lipid:drug ratio of about 10:1.

#4.3 Experiment Formulation:

2 mg amikacin in 1.0 mL D.D water was filtered through 0.22 μ m filter and combined with 20 mg of 50% soy PS in 2.0 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 0.206 mL of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 76 mg sodium chloride in 264 μ L sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the amikacin at 0.5 mg/mL with sterile water. Also, the final product containing 0.33M sodium chloride. To make nice crystalline cochleates of amikacin with a lipid:drug ratio of about 10:1.

#4.4 Experiment Formulation:

2 mg amikacin in 1.0 mL D.D water was filtered through 0.22 μ m filter and combined with 20 mg of 50% soy PS in 2.0 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 0.206 mL of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 152 mg sodium chloride in 524 μ L sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the amikacin at 0.5 mg/mL with sterile water. Also, the final product containing 0.66M sodium chloride. To make nice crystalline cochleates of amikacin with a lipid:drug ratio of about 10:1.

Example 5 Amikacin Crystal Cochleate Formulation for In Vivo Study

Procedure of Amikacin Formulation for In Vivo Study:

#5.1 Experiment Formulation:

200 mg amikacin in 20 mL D.D water was filtered through 0.22 μ m filter and combined with 2000 mg of 50% soy PS in 200 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 17 mL of 0.1M calcium chloride to form cochleates. Cochleates were then concentrated under lyophilization to provide crystal cochleates (about 6.7 mg/mL) with a lipid:amikacin ratio of about 10:1.

#5.2 Experiment Formulation:

200 mg amikacin in 20 mL D.D water was filtered through 0.22 μ m filter and combined with 2000 mg of 50% soy PS in 200 mL sterile water (liposome was filtered through 5, 0.8 and 0.45 μ m filter) to form liposomes containing the amikacin. To the resultant mixture was then added 17 mL of 0.1M calcium chloride to form cochleates. To reduce the

aggregation size of crystals cochleates, 1125.2 mg sodium chloride in 3.88 mL sterile water was then added to the mixture of the cochleates. Cochleates were then concentrated under lyophilization to provide crystal cochleates (about 6.7 mg/mL, and 0.66M sodium chloride) with a lipid:amikacin ratio of about 10:1.

Example 6 In Vitro Study of Aminoglycosides Crystal Cochleate Formulation

Macrophages and Infection

Cell lines: mouse peritoneal macrophage cell line (Raw 246.7), and/or THP-1, a human macrophage cell line. Cells were cultured in DMEM and RPMI-1640, respectively, supplemented with 5% heat-inactivated fetal bovine serum. Macrophage monolayers were established by adding 10^5 macrophages to a 24-well tissue culture plate. After 24 hours, monolayers were infected and the infection was allowed to happen for 1 hour, and then the extracellular bacteria were removed by washing. Some of the well contents were lysed and plated onto Middlebrook 7H10 agar plate, to determine the intracellular inoculum of the bacterium. The remaining wells were treated daily with different aminoglycosides (i.e., amikacin, gentamicin, and paromomycin) cochleates prepared in accordance with the claimed method. After treatment, cell monolayers were lysed and the lysate plated into 7H10 agar to quantify the intracellular load.

TABLE 1

Efficacy Comparison between Encochleated Aminoglycosides Formulation and Free Aminoglycosides	
Results	Enhanced Efficacy vs. Free Drug
Amikacin	<i>Mycobacterium avium</i> (10x-20x) <i>Mycobacterium tuberculosis</i> (7x) <i>Francisella tularensis</i> LVS (3x)
Gentamicin	<i>Mycobacterium avium</i> (10x) <i>Mycobacterium smegmatis</i> (50x) <i>Mycobacterium tuberculosis</i> (2x) <i>Francisella tularensis</i> LVS (2x)
Paromomycin	<i>Francisella tularensis</i> type A (4x) <i>Francisella tularensis</i> type A (4x) Cutaneous Leishmaniasis (3x)

As indicated in Table 1, aminoglycoside cochleates have enhanced efficacy against different bacteria compared to corresponding non-encocheated free drugs.

Example 7 In Vitro Study of Amikacin Crystal Cochleate Formulation

Methods: As shown in Example 4, amikacin cochleate (Amkcch) formulations were optimized for amikacin encocheation efficiency and particle size by varying the type of PS used, the PS: biological active ratio, PS: Ca^{++} ratio, and NaCl concentration. The efficacy of Amkcch against intracellular Ma infections was evaluated in vitro using mouse peritoneal macrophage infected with *M. avium* strains MAC 101 or MAC 109. Mouse peritoneal macrophages (Mo) Raw 264.7 cells were seeded at 105 cells/well. Mo monolayers were infected at ratio 1:10 for 1 h and extracellular bacteria removed. Monolayers were treated with free amikacin and/or cochleate preparations for 4 days and the number of intracellular bacteria determined. Assays were repeated three times.

Results: Untreated control Ma strains grew within Mo from 3.8' 105 to 4.9'106. Ma within Mo treated with free

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amikacin (10 and 20 mg/mL) were killed to 6.1 and 3.4' 10⁴ bacteria, respectively. Optimized Amkcch (10 and 20 mg/mL) demonstrated greater than 10-fold enhanced efficacy, reducing bacterial count to 3.9 and 1.7' 10³ bacteria within Mo (p<0.05 compared with free amikacin).

Conclusions: As indicated in the FIG. 1, amikacin cochleate formulations are 10-50 fold more active than free amikacin against *M. avium* infection in macrophage. Cochleate preparations of amikacin showed significant and enhanced activity against Ma strains in macrophages, suggesting that cochleates achieved higher intracellular concentration for a longer time than free amikacin.

Example 8 In Vivo Study of Amikacin Crystal Cochleate Formulation

A formation of amikacin cochleates has been developed. The in vivo efficacy amikacin cochleats against *Mycobacterium avium* complex (MAC) was evaluated using C57BL/6 black mice.

Mice, 12/group, were infected with *M. avium* 101 (8.1×10⁷ bacterial/mouse) by tail vein injection.

After 7 days, 6 mice were harvested an the number of MAC in spleen was quantified to establish the baseline bacterial load (Time 0).

Mice were treated with various amikacin cochleates as indicated in FIGS. 2 and 3

at 1.0 mg amikacin/day for 2 weeks.

Mice were harvested at week 3 and 2 days later (after 2 weeks of treatment), and spleens homogenized and plated onto 7H10 agar.

Colonies on plates were counted and the data were analyzed.

As demonstrated in FIGS. 2 and 3, amikacin cochleates, given I.P. or orally, were active, reducing the number of bacterial load in the spleen. The amikacin cochleate preparation with high salt concentration does orally was

Conclusion: Oral delivery of Amikacin cochleate formulations demonstrates in vivo efficacy similar to IP free amikacin.

Example 9 High Salt Procedure for Aminoglycoside Formulations for In Vitro Study

#9.1 Experiment Formulation:

2 mg aminoglycoside in 1.0 ml D.D water was filtered through 0.22 µm filter and combined with 20 mg of 50% soy PS in 2.0 ml sterile water (liposome was filtered through 5, 0.8 and 0.45 µm filter) to form liposomes containing the

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aminoglycoside. To the resultant mixture was then added 0.206 ml of 0.1M calcium chloride to form cochleates. The mixture was then adjusting the drug concentration of the aminoglycoside at 0.5 mg/ml with sterile water. To make nice crystalline cochleates of aminoglycoside with a lipid: drug ratio of about 10:1.

#9.2 Experiment Formulation:

2 mg aminoglycoside in 1.0 ml D.D water was filtered through 0.22 µm filter and combined with 20 mg of 50% soy PS in 2.0 ml sterile water (liposome was filtered through 5, 0.8 and 0.45 µm filter) to form liposomes containing the aminoglycoside. To the resultant mixture was then added 0.206 ml of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 15 mg sodium chloride in 52 µl sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the aminoglycoside at 0.5 mg/ml with sterile water. Also, the final product containing the 0.066M sodium chloride. To make nice crystalline cochleates of aminoglycoside with a lipid:drug ratio of about 10:1.

#9.3 Experiment Formulation:

2 mg aminoglycoside in 1.0 ml D.D water was filtered through 0.22 µm filter and combined with 20 mg of 50% soy PS in 2.0 ml sterile water (liposome was filtered through 5, 0.8 and 0.45 µm filter) to form liposomes containing the aminoglycoside. To the resultant mixture was then added 0.206 ml of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 76 mg sodium chloride in 264 µl sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the aminoglycoside at 0.5 mg/ml with sterile water. Also, the final product containing 0.33M sodium chloride. To make nice crystalline cochleates of aminoglycoside with a lipid:drug ratio of about 10:1.

#9.4 Experiment Formulation:

2 mg aminoglycoside in 1.0 ml D.D water was filtered through 0.22 µm filter and combined with 20 mg of 50% soy PS in 2.0 ml sterile water (liposome was filtered through 5, 0.8 and 0.45 µm filter) to form liposomes containing the aminoglycoside. To the resultant mixture was then added 0.206 ml of 0.1M calcium chloride to form cochleates. To reduce the aggregation size of the crystals cochleates, 152 mg sodium chloride in 524 µl sterile water was then added to the mixture of the cochleates. The mixture was then adjusting the concentration of the aminoglycoside at 0.5 mg/ml with sterile water. Also, the final product containing 0.66M sodium chloride. To make nice crystalline cochleates of aminoglycoside with a lipid:drug ratio of about 10:1.

TABLE 2

Results of Amikacin Encochleation for in House Study Obtained from Amikacin-cochleation Supernatant (Degussa 85% vs Lipoid 50% Soy PS and before and after lyophilization)					
Measured by using	Ratio of the Lipid:Drug	Deguss 85% bSoy PS before lyophilization	Deguss 85% bSoy PS After lyophilization	Lipoid 50% Soy PS before lyophilization	Lipoid 50% Soy PS After lyophilization
Buffer	20:1	34% in sup	39.5% in sup	34.5% in sup	33.8% in sup
Buffer	10:1	69.1% in sup	72.6% in sup	50% in sup	48.3% in sup
Buffer	5:1	74.7% in sup	83.5% in sup	61.3% in sup	58.1% in sup
Degussa 85% soy PS Plain Sup	20:1	25.3% in sup	30.2% in sup	25.7% in sup	25% in sup
Degussa 85% soy PS Plain Sup	10:1	55.4% in sup	58.4% in sup	38.4% in sup	36.8% in sup
Degussa 85% soy PS Plain Sup	5:1	61.6% in sup	69.5% in sup	49.5% in sup	47% in sup
Lipoid 50% Soy PS Plain Sup	20:1	29.6% in sup	34.3% in sup	30% in sup	29.5% in sup

TABLE 2-continued

Results of Amikacin Encochleation for in House Study Obtained from Amikacin-cochleation Supernatant (Degussa 85% vs Lipoid 50% Soy PS and before and after lyophilization)					
Measured by using	Ratio of the Lipid:Drug	Deguss 85% bSoy PS before lyophilization	Deguss 85% bSoy PS After lyophilization	Lipoid 50% Soy PS before lyophilization	Lipoid 50% Soy PS After lyophilization
Lipoid 50% Soy PS Plain Sup	10:1	59.8% in sup	62.9% in sup	43.6% in sup	42.4% in sup
Lipoid 50% Soy PS Plain Sup	5:1	64.3% in sup	72.2% in sup	52.8% in sup	50.8% in sup

Encochleation percentage is determined by measuring the amount of amikacin in the supernatant (ninhydrin assay) after forming the cochleates and then centrifuging. The amount in the supernatant is then subtracted from the total amount during the encochleation process. This is reported as percent in supt. The lower the percent in supernatant the higher the encochleation efficiency. The results listed in Table 2 clearly demonstrate that 50% soy PS is more efficient at encochleating than 85% PS.

higher the encochleation efficiency. The results listed in Table 4 clearly demonstrate that 50% soy PS is more efficient at encochleating than 99.99% PS.

FIG. 4 shows Gentamicin Encochleation Related to PS Properties. Similar to Tables 1-3, the results listed in FIG. 4 clearly demonstrate that 50% soy PS is more efficient at encochleating gentamicin at lower bile salt concentrations than 85% PS or 99.99% PS.

TABLE 3

Results of Amikacin Formulation Obtained from Absorbance at 400 nm						
Measured by Using	Ratio of 50% soy PS to drug (10:1)	Ratio of 85% soy PS to drug (10:1)	Ratio of 50% soy PS to drug (20:1)	Ratio of 85% soy PS to drug (20:1)	Ratio of 50% soy PS to drug (5:1)	Ratio of 85% soy PS to drug (5:1)
50 mM Phosphate buffer + Amikacin	50.6% in sup	94.4% in sup	34.8% in sup	54.9% in sup	53.5% in sup	76.4% in sup
50% soyPS Plain sup + Amikacin	41% in sup	79% in sup	27.9% in sup	45.2% in sup	44.3% in sup	63.8% in sup
85% soyPS Plain sup + Amikacin	44% in sup	82.8% in sup	30% in sup	47.8% in sup	46.8% in sup	66.9% in sup

Encochleation percentage is determined by measuring the amount of amikacin in the supernatant (ninhydrin assay) after forming the cochleates and then centrifuging. The amount in the supernatant is then subtracted from the total amount during the encochleation process. This is reported as percent in supt. The lower the percent in supernatant the higher the encochleation efficiency. The results listed in Table 3 clearly demonstrate that 50% soy PS is more efficient at encochleating than 85% PS.

Example 10 Encochleation Efficiency of Amikacin, Gentamicin, and Paromomycin Bile Salt Cochleates

#10.1 Procedure for Aminoglycoside Formulation with Bile Salts (after Calcium) for In Vitro Studies.

Crystalline cochleates of aminoglycoside formulation with different amount of bile salts (Sigma—Aldrich catalog #48305 Fluka cholic acid sodium salt, ~50% deoxycholic acid sodium salt, ~50%):

TABLE 4

Results of Amikacin Formulation Obtained from Absorbance at 400 nm					
Measured by Using	Ratio of DOPS to Amikacin	Lipoid 50% Soy PS	Avanti DOPS	Avanti 99% Soy PS	NOF DOPS
50 mM Phosphate buffer	10:1	47.3% in sup	87.7% in sup	84.3% in sup	86.4% in sup
Lipoid 50% soy PS Plain cochleates sup	10:1	37.1% in sup	77% in sup	73.5% in sup	64% in sup
Avanti DOPS plain sup	10:1	43.6% in sup	83% in sup	80% in sup	81.7% in sup
Avanti 99% soy PS plain sup	10:1	41.9% in sup	84.6% in sup	81% in sup	83% in sup

Encochleation percentage is determined by measuring the amount of amikacin in the supernatant (ninhydrin assay) after forming the cochleates and then centrifuging. The amount in the supernatant is then subtracted from the total amount during the encochleation process. This is reported as percent in supt. The lower the percent in supernatant the

Aminoglycoside, 2 mg in 0.2 ml sterile water was filtered through a 0.22 μ m filter and combined with 20 mg of 50% soy PS liposome in 2.0 ml sterile water (the soy PS liposome was first filtered through 5, 0.8 and 0.45 μ m filters) to form liposomes containing the aminoglycoside. To the resultant mixture, 0.159 ml of 0.1M calcium chloride was then added

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Enochleation percentage is determined by measuring the amount of aminoglycoside bile salts in the supernatant (ninhydrin assay) after forming the cochleates and then centrifuging. The amount in the supernatant is then subtracted from the total amount during the enochleation process. These results clearly demonstrate that 50% soy PS is more efficient at enochleating at lower bile salt concentrations than 75% PS or 99.99% PS. A is bile salts added after the enochleation. B is bile salts added before the enochleation.

What is claimed is:

1. A method of treating a subject with a fungal infection comprising, administering to the subject a composition comprising a cochleate comprising (i) a soy-based phospholipid comprising about 40%-55% by weight soy phosphatidylserine, (ii) a multivalent cation, and (iii) an anti-fungal agent.

2. The method of claim 1 wherein the soy-based phospholipid comprises about 45%-55% by weight soy phosphatidylserine.

3. The method of claim 1, wherein the soy-based phospholipid comprises about 40% by weight soy phosphatidylserine.

4. The method of claim 1, wherein the soy-based phospholipid comprises about 45% by weight soy phosphatidylserine.

5. The method of claim 1, wherein the soy-based phospholipid comprises about 50% by weight soy phosphatidylserine.

6. The method of claim 1, wherein the soy-based phospholipid comprises about 55% by weight soy phosphatidylserine.

7. The method of claim 1, wherein the antifungal agent is amphotericin B.

8. The method of claim 1, wherein the multivalent cation is Ca^{++} , Zn^{++} , Ba^{++} , or Mg^{++} .

9. The method of claim 8, wherein the multivalent cation is Ca^{++} .

10. The method of claim 1, wherein the soy-based phospholipid is a mixture comprising (i) soy phosphatidylserine and (ii) phosphatidylcholine, phosphatidylinositol, phosphatidylglycerol, or any combination thereof.

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11. The method of claim 1, wherein the soy-based phospholipid is a mixture comprising (i) soy phosphatidylserine and (ii) phosphatidylcholine, phosphatidylinositol, or phosphatidylglycerol.

12. The method of claim 11, wherein the mixture comprises (i) soy phosphatidylserine and (ii) phosphatidylcholine.

13. The method of claim 1, wherein the subject is immune compromised.

14. The method of claim 1, wherein the composition is orally administered to the subject.

15. A method of treating a subject with a fungal infection comprising, administering to the subject a composition comprising a cochleate comprising (i) a soy-based phospholipid comprising a mixture comprising soy phosphatidylserine and phosphatidylcholine, (ii) a multivalent cation, and (iii) amphotericin B, wherein the soy-based phospholipid comprises about 40%-55% by weight soy phosphatidylserine.

16. The method of claim 15 wherein the soy-based phospholipid comprises about 45%-55% by weight soy phosphatidylserine.

17. The method of claim 15, wherein the soy-based phospholipid comprises about 40% by weight soy phosphatidylserine.

18. The method of claim 15, wherein the soy-based phospholipid comprises about 45% by weight soy phosphatidylserine.

19. The method of claim 15, wherein the soy-based phospholipid comprises about 50% by weight soy phosphatidylserine.

20. The method of claim 15, wherein the soy-based phospholipid comprises about 55% by weight soy phosphatidylserine.

21. The method of claim 15, wherein the multivalent cation is Ca^{++} , Zn^{++} , Ba^{++} , or Mg^{++} .

22. The method of claim 21, wherein the multivalent cation is Ca^{++} .

23. The method of claim 15, wherein the subject is immune compromised.

24. The method of claim 15, wherein the composition is orally administered to the subject.

* * * * *